

Importing Required Libraries

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Team ID	
Project Name	A Comprehensive Measure of Well-Being

Required Libraries Install and Import

```
✓ import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

import pickle

%matplotlib inline
✓ 37.5s
```

Description

All the required Python libraries and modules are imported to support data analysis, preprocessing, machine learning model development, evaluation, and web application deployment. Pandas is used for loading, organizing, and manipulating the dataset, while NumPy performs numerical and mathematical operations efficiently. Matplotlib and Seaborn are imported to create visualizations such as graphs and plots that help analyze the distribution and relationships within the dataset. From Scikit-learn, the LabelEncoder is used to convert categorical values into numerical format, train_test_split divides the dataset into training and testing sets, LinearRegression is used to train the prediction model, and r2_score evaluates the model's performance. The Pickle library is used to save the trained model so that it can be loaded later in the Flask application without retraining. Finally, %matplotlib inline enables the generated plots to be displayed directly within the Jupyter Notebook during analysis and model development.